



ID	X	Y	DIA	KEEPOUT
H1	3.50	3.50	2.75 +/-0.05	not given
H2	61.50	3.50	2.75 +/-0.05	not given
H3	3.50	52.50	2.75 +/-0.05	not given
H4	61.50	52.50	2.75 +/-0.05	not given

#	FEATURE	X EXTENT	Y EXTENT
1	40-pin GPIO header	7.10 to 57.90	50.00 to 55.00
2	Ethernet RJ45	65.65 to 87.00	2.50 to 18.00
3	USB 2.0 type A (upper pair)	69.30 to 87.00	39.65 to 53.57
4	USB 2.0 type A (lower pair)	69.30 to 87.00	21.65 to 35.57
5	micro-USB power input	6.85 to 14.35	-0.60 to 4.71
6	HDMI type A	24.75 to 39.25	-1.50 to 10.65
7	3.5 mm A/V jack	50.00 to 57.00	0.00 to 12.50

PORT	EDGE	CX	CY
JA	left edge	9.40	39.29
JB	left edge	9.40	16.43
JC	bottom edge	27.60	9.40

1. All dimensions in millimetres. Datum is the lower-left corner of the board outline; X to the right, Y up, viewed from the component side.
2. Hole and Pmod positions are ordinate dimensions from that single datum, so they do not accumulate tolerance.
3. Phantom outline is the Digilent Pmod HAT Adapter fitted to the 40-pin GPIO header. Its Pmod host positions are given in this board's coordinate frame, since the adapter's mounting holes coincide with this board's.
4. Connector outlines are the component body as drawn by Raspberry Pi Ltd, including any overhang past the board edge.
5. Mounting hole diameter: Drawing note, quoted: 4x M2.5 MOUNTING HOLES DRILLED TO 2.75 +/- 0.05mm.
6. No keep-out around the mounting holes is dimensioned on this model's own drawing. The Raspberry Pi HAT mechanical specification ([github.com/raspberrypi/hats, hat-board-mechanical.pdf](https://github.com/raspberrypi/hats/blob/master/hat-board-mechanical.pdf)) requires a 6.2 mm keep-out around each mounting hole on a board fitted to a Pi; design to that.
7. Hole positions are the 3.5 mm inset and 58 x 49 mm rectangle dimensioned on this model's own drawing.
8. Pmod host JC on the Pmod HAT Adapter overhangs the lower board edge in the same region as this board's HDMI and power connectors. Digilent's own reference manual warns to fit the two standoffs opposite the 40-pin connector so that JC's pins cannot touch the HDMI shell.
9. Pmod HAT Adapter: DERIVED DIMENSIONS. Digilent publish no mechanical drawing, DXF, STEP or board file for this product. Pmod host positions were measured photographically and are good to about +/-0.75 mm.
10. Pmod HAT Adapter: JA and JB face out of the left edge, JC out of the lower edge. All three are right-angle hosts: the connector body overhangs the board edge by roughly 3 mm and the through-hole field sits about 9.4 mm in from it, so a mating Pmod peripheral plugs

11. **Preorder HAT Assembly:** <https://datasheets.raspberrypi.com/raspberry-pi/b-ordering-guide.pdf> states that the 23.86mm pins spacing of the Pi Mechatronics board is significant. Raspberry Pi the 23.18 mm apart, inside the measurement uncertainty.
12. **Mounting holes:** <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.pdf> - Drawing note, quoted: 4x M2.5
12. **ROUTING HOLE DIMENSIONS:** <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.pdf> - Drawing note, quoted: 4x M2.5

85 x 56 mm

DRAWN	DATE	UNITS	SCALE
generated	2026-09-10	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPI-02	A